

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF CYBER SECURITY LANDSCAPE

In the modern digital era, cyber security has become one of the most critical concerns for individuals, organizations, and governments worldwide, as the continuous evolution of technology has significantly expanded the attack surface for cyber threats, and by 2026, the cyber security landscape is expected to grow even more complex due to the rapid adoption of cloud computing, the Internet of Things (IoT), artificial intelligence, and highly interconnected digital systems.

Cyber threats such as phishing, ransomware, malware injection, data breaches, and identity theft are increasing at an alarming rate, with attackers leveraging advanced technologies, automation tools, and AI to execute sophisticated attacks that are difficult to detect using traditional systems, while organizations struggle to keep pace with these evolving threats, making real-time detection and response a major challenge.

A key limitation of existing cyber security systems is their reliance on visual monitoring through dashboards, notifications, and logs that require

constant human attention, which in practical environments like laboratories, offices, or monitoring centers often leads to delayed responses.

Additionally, with the rise of IoT devices and smart environments, there is an increasing need for integrated and automated security solutions, as cyber security now extends beyond software alone and requires a combination of intelligent software and hardware-based alert mechanisms to ensure immediate awareness and rapid response.

1.2 THE RISK OF DARK WEB VULNERABILITIES

The Dark Web has emerged as one of the most significant contributors to the growing landscape of modern cyber security threats. Unlike the surface web, which is indexed and accessible through standard search engines, the Dark Web operates on encrypted and anonymized networks such as the Tor (The Onion Router) network, allowing users to communicate and transact while maintaining near-complete anonymity. This anonymity has fostered the rapid growth of illegal and malicious activities, transforming the Dark Web into a thriving underground ecosystem where cybercriminals freely exchange stolen credentials, financial records, corporate data, exploit kits, malware tools, and zero-day vulnerabilities.

A significant portion of modern cybercriminal activity is coordinated through Dark Web forums, marketplaces, and leak platforms. On these platforms, attackers share system vulnerabilities, advertise hacking services,

distribute ransomware, and plan large-scale cyberattacks well before they are executed on the surface web. These environments act as early-stage breeding grounds for threats, providing malicious actors with the resources, knowledge, and coordination necessary to conduct sophisticated and targeted attacks against individuals, organizations, and governments.

The information exchanged often contains early indicators of malicious intent, making Dark Web monitoring a critical component of proactive threat detection. Therefore, integrating Dark Web scraping, monitoring, and intelligent analysis into modern cyber security frameworks has become essential for early threat identification.

1.3 PROBLEM STATEMENT

Despite advancements in cyber security technologies, several limitations still exist in current systems:

- ❖ Most systems rely only on software-based alerts and require continuous user monitoring.
- ❖ Lack of real-time physical alert mechanisms to notify users instantly.
- ❖ Limited capability to analyse dark web content and hidden threats.
- ❖ Delayed response due to dependency on dashboards and notifications.

❖ Absence of integration between web-based detection and IoT-based alert Systems.

1.4 SUSTAINABLE DEVELOPMENT GOAL (SDG) MAPPING

The proposed Dark Web Threat Intelligence Scraper and Automated Threat Classification System aligns with the objectives of the United Nations Sustainable Development Goals 4 (Quality Education) and 16 (Peace, Justice and Strong Institutions) as shown in Figure 1.1.



Figure 1.1 Supported Sustainable Development Goals

The Dark Web Threat Intelligence Scraper helps identify leaked educational data, phishing links, malicious forums, and cyber threats targeting students, researchers, and academic institutions. The Automated

Threat Classification using Machine Learning improves cyber security awareness and digital literacy by automatically detecting harmful online activities. Educational organizations can use this system to protect online learning platforms, student databases, and research resources from cyberattacks. By ensuring safer digital education systems, the project contributes to uninterrupted and quality learning experiences. The system monitors dark web activities to detect illegal cyber threats, data leaks, fraud discussions, and malicious activities. Machine learning models classify threats automatically, helping authorities and organizations respond quickly to cyber risks.

1.5 COMPLEX ENGINEERING PROBLEM ADDRESSED

The proposed system addresses the complex engineering problem of identifying and classifying cyber threats from anonymous dark web sources using automated Machine Learning techniques. The major challenge lies in collecting hidden dark web data through TOR networks, processing large-scale unstructured threat information, and accurately detecting malicious activities in real time. Traditional security systems are often unable to efficiently monitor evolving cyber threats such as phishing attacks, malware distribution, credential leaks, and illegal communications present in dark web platforms.

To overcome these challenges, the system integrates dark web scraping, Natural Language Processing (NLP), and Machine Learning algorithms to automate threat intelligence collection and threat classification. The project

enhances proactive cyber security by enabling faster threat detection, reducing manual analysis efforts, improving classification accuracy, and supporting secure digital infrastructures against emerging cyber threats.

The project also addresses the challenge of developing an intelligent and scalable cyber security framework capable of adapting to continuously evolving dark web activities. Since cybercriminals frequently change communication patterns, attack methods, and hidden platforms, manual monitoring becomes inefficient and time-consuming. The proposed system utilizes automated data extraction, feature analysis, and predictive Machine Learning models to improve the efficiency of cyber threat monitoring and decision-making.

1.6 OBJECTIVES OF THE PROJECT

The primary objectives of this project are centered around building a comprehensive, intelligent, and real-time cyber threat detection system that bridges the gap between digital threat intelligence and physical alert mechanisms. The system is designed to analyze URLs from both the surface web and the Dark Web, enabling a broader and more complete scope of threat visibility compared to conventional cyber security tools.

To ensure immediate and reliable user awareness, the system implements real-time IoT-based alert mechanisms using visual indicators

such as LEDs, an audio buzzer, and an LCD display, which provide physical notifications even when the user is not actively monitoring the screen.

A user-friendly web application and browser extension are developed to make the system accessible and practical for everyday use, while Telegram notifications are integrated to deliver instant remote alerts to users regardless of their location. A user-friendly web application and browser extension are developed to make the system accessible and practical for everyday use, while Telegram notifications are integrated to deliver instant remote alerts to users regardless of their location.

1.7 MOTIVATION AND SCOPE

The growing sophistication of cyber threats in the modern digital era exposed critical limitations in traditional cyber security systems, which predominantly operate in a reactive manner — detecting and responding to attacks only after a system has already been compromised.

As highlighted in the IEEE paper on proactive cyber threat detection, a significant portion of cyberattack planning, including vulnerability discussions, malware trading, and ransomware coordination, takes place on the Dark Web well before execution, making early detection an urgent necessity. Despite this, most existing cyber security solutions lack the capability to monitor, access, or analyze Dark Web content effectively, leaving organizations with a dangerous blind spot in their threat intelligence

pipeline. In real-world environments such as laboratories, offices, and security monitoring centers, users may not always be actively observing dashboards or notification panels, resulting in missed alerts and delayed responses.

This creates a strong motivation for integrating physical, IoT-based alert mechanisms into the cyber security pipeline, ensuring that threats are communicated immediately and reliably through hardware indicators such as LEDs, buzzers, and LCD displays, regardless of whether the user is monitoring the interface.

The proactive cyber threat detection framework described in the IEEE paper, the system is designed to continuously collect and analyze threat-related data from both surface web and Dark Web sources, employing NLP techniques and machine learning classifiers to detect early indicators of malicious activity and assess threat severity.

1.8 EVOLUTION OF CYBER THREATS IN THE DIGITAL ERA

The digital era has significantly transformed cyber threats from simple virus attacks to highly organized and sophisticated cybercrimes targeting critical infrastructures, financial systems, healthcare sectors, and government networks. With the rapid growth of technologies such as cloud computing, IoT, and artificial intelligence, cybercriminals increasingly use

Dark Web platforms to plan and execute advanced threats including ransomware, phishing, supply chain attacks, and zero-day exploits.

The evolution of cyber threats in the digital era has been driven by rapid technological advancements and the increasing dependence on 6 interconnected systems. In the early stages of computing, cyber threats were relatively simple, often limited to basic viruses and unauthorized access attempts.

1.9 ROLE OF ARTIFICIAL INTELLIGENCE IN MODERN CYBER SECURITY

Artificial Intelligence (AI) plays a crucial role in modern cyber security by enhancing the ability to detect, prevent, and respond to increasingly sophisticated cyber threats. Traditional security systems often rely on predefined rules and signatures, which are ineffective against new and evolving attacks.

In contrast, AI-driven systems use machine learning algorithms to analyze large volumes of data, identify patterns, and detect anomalies in real time, enabling early identification of potential threats such as malware, phishing, and intrusion attempts. AI also improves threat intelligence by continuously learning from past incidents and adapting to emerging attack techniques.

Artificial Intelligence also plays a significant role in predictive threat analysis, where it anticipates potential cyber attacks before they occur by analysing historical data and emerging patterns. This proactive approach helps organizations strengthen their defenses in advance rather than reacting after an attack has taken place.

AI is widely used in intrusion detection and prevention systems (IDS/IPS), where it continuously monitors network traffic and flags unusual behaviour that may indicate a security breach.

1.10 SIGNIFICANCE OF PROACTIVE VS REACTIVE SECURITY

The significance of proactive versus reactive security lies in how effectively an organization can prevent and respond to cyber threats. Reactive security focuses on responding to attacks after they occur, involving actions such as incident response, damage control, and system recovery.

While this approach is necessary, it often leads to data loss, financial damage, and reputational harm because the threat has already impacted the system. In contrast, proactive security emphasizes anticipating and preventing threats before they happen by using techniques such as continuous monitoring, vulnerability assessment, threat intelligence, and predictive analysis. Reactive security is a cyber security approach that focuses on responding to threats after they have occurred. It mainly deals

with identifying attacks, minimizing damage, recovering affected systems, and restoring normal operations.

Traditional security systems such as antivirus software, firewalls, and incident response mechanisms operate reactively by taking action only after suspicious or malicious activity is detected. In the context of dark web threat intelligence, reactive security helps organizations analyze cyber incidents such as data breaches, phishing attacks, malware infections, and unauthorized access after the attack has already impacted the system.

1.11 TECHNOLOGY READINESS LEVEL (TRL) ASSESSMENT

The proposed project, Dark Web Threat Intelligence Scraper and Automated Threat Classification using Machine Learning, is currently positioned at Technology Readiness Level (TRL) 5–6. The system has been developed with integrated components including dark web scraping, Natural Language Processing (NLP), Machine Learning-based threat classification, and automated threat analysis. Individual modules have been tested in a simulated and controlled development environment to validate their functionality and performance.

The project demonstrates the feasibility of collecting dark web data through TOR-based scraping techniques and classifying cyber threats using trained Machine Learning models. Core functionalities such as data preprocessing, feature extraction, threat prediction, and automated analysis

have been successfully implemented and evaluated with sample datasets. The frontend and backend integration further supports real-time interaction and automated processing capabilities. The system is capable of supporting cyber security monitoring applications in educational institutions, organizations, and digital platforms. However, large-scale deployment, real-time industrial testing, and continuous model optimization are required to achieve higher TRL levels.

Future enhancements such as cloud deployment, advanced threat intelligence integration, and improved scalability can further increase the system's readiness for real-world cyber security environments. Developed a functional prototype for dark web threat intelligence scraping and automated threat classification. Integrated TOR-based dark web data collection and monitoring mechanisms. Requires large-scale real-world deployment and continuous optimization for higher TRL advancement.

CHAPTER 2

LITERATURE REVIEW

2.1 ANALYSIS OF RNN-LSTM IN THREAT DETECTION

Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models have become highly significant in threat detection due to their ability to process sequential data and identify patterns over time. Traditional machine learning techniques often struggle with time-dependent data, whereas RNNs are specifically designed to analyze sequences such as network traffic, user behavior, and textual data from web sources.

However, standard RNNs face limitations like vanishing and exploding gradient problems, which reduce their effectiveness in capturing long-term dependencies. To overcome this, LSTM networks were introduced with specialized memory cells and gating mechanisms—input, output, and forget gates—that allow the model to retain important information while discarding irrelevant data.

In cyber security, RNN-LSTM models are widely used for applications such as intrusion detection systems (IDS), malware detection, phishing identification, and anomaly detection. They can analyze sequences of events,

such as login attempts or packet flows, to detect unusual patterns that may indicate a cyberattack.

For example, an LSTM model can identify suspicious behavior in a series of network requests or detect malicious keywords and structures in URLs and web content. Compared to traditional approaches, LSTM-based models often achieve higher accuracy because they can learn complex temporal relationships in data.

2.2 SURVEY OF TOR AND ANONYMITY NETWORKS

The Tor (The Onion Router) network is one of the most widely used anonymity networks for accessing the hidden parts of the internet, commonly known as the dark web. It enables users to browse and communicate anonymously by routing internet traffic through multiple encrypted layers and relay nodes, making it extremely difficult to trace the origin or destination of the data. This decentralized architecture provides enhanced privacy and security for users, but it is also widely exploited by cybercriminals for illegal activities such as malware distribution, data leaks, phishing operations, and anonymous communication of cyber threats.

Each layer of encryption is decrypted at different nodes in the network, ensuring that no single point has complete information about the communication path. This mechanism provides strong privacy protection and is widely used for legitimate purposes such as secure communication,

protecting user identity, and bypassing censorship. However, the same anonymity features that make Tor valuable for privacy also make it attractive for cybercriminal activities. Dark web platforms operating on Tor are often used for illegal purposes such as selling stolen data, distributing malware, offering hacking services, and conducting illicit transactions. These platforms rely on anonymity to evade law enforcement and continue operations. As a result, Tor has become a significant area of concern in cyber security, as many cyber threats originate or are coordinated through such hidden networks. Figure 2.1 illustrates the TOR/ONION page content extraction process of proposed system.

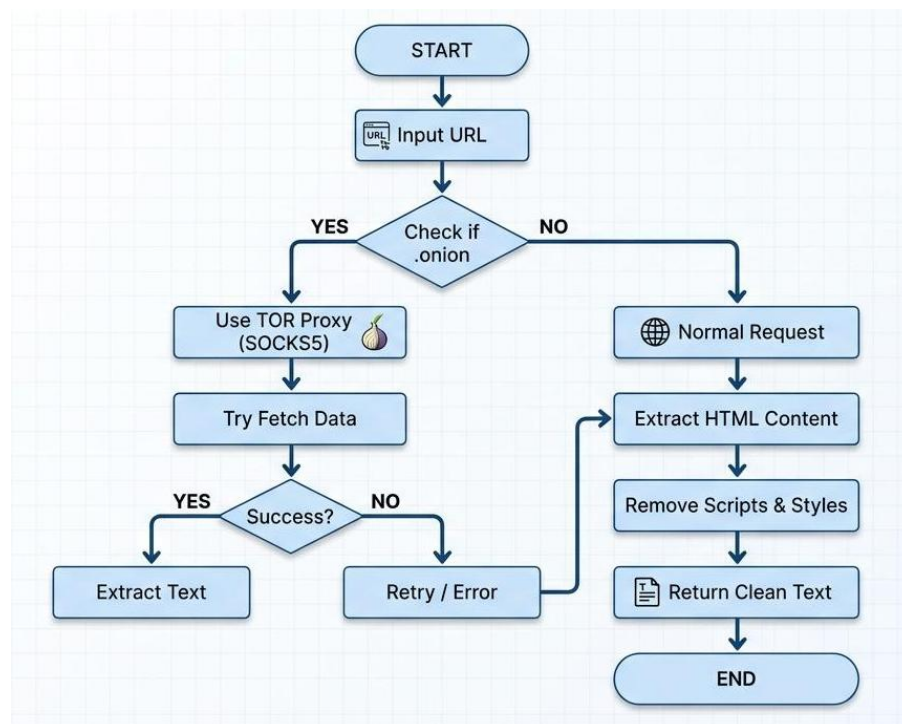


Figure 2.1 TOR/ONION page content extraction process

2.3 EXISTING IOT ALERT SYSTEMS: A COMPARATIVE STUDY

The Internet of Things has revolutionized the way physical environments interact with digital systems, enabling real-time monitoring and automated responses across a wide range of domains including home automation, industrial monitoring, healthcare, and environmental sensing.

IoT-based alert systems typically consist of microcontrollers, sensors, communication modules, and output devices such as LEDs, buzzers, and display units that work together to detect events and notify users instantly through physical indicators. Table 2.1 illustrates the Comparative Analysis Of Existing Systems of proposed system.

Table 2.1 Comparative Analysis of Existing System

Feature	Traditional Cybersecurity	IoT Alert Systems	Proposed System
Alert Type	Visual (screen)	Physical (LED/Buzzer)	Both
Real-time Response	Medium	High	High
User Dependency	High	Low	Low
Dark Web Analysis	No	No	Yes

This integration represents a significant advancement over existing solutions, ensuring that users receive instant, reliable notifications regardless

of whether they are actively monitoring their screens, and positioning the proposed system as a more complete and practical cyber security solution for real-world deployment.

2.4 NATURAL LANGUAGE PROCESSING IN CYBER SECURITY

Natural Language Processing has emerged as a transformative technology in the field of cyber security, enabling automated systems to extract meaningful threat intelligence from the vast volumes of unstructured textual data generated across online platforms, forums, social media, and Dark Web sources.

Unlike structured data, which follows predefined formats, the textual content found in cybercriminal communications is highly informal, coded, and context-dependent, making traditional rule-based analysis insufficient for accurate threat detection.

NLP techniques such as named entity recognition, topic modeling, sentiment analysis, keyword extraction, and contextual semantic analysis allow cyber security systems to identify threat-related patterns, detect malicious intent, and classify content with a level of accuracy and scalability that manual analysis cannot achieve. In the context of Dark Web intelligence, NLP plays a particularly critical role, as underground forums and marketplaces predominantly communicate through unstructured text filled

with slang, abbreviations, aliases, and obfuscated language specifically designed to evade detection.

2.5 MACHINE LEARNING APPROACHES FOR THREAT CLASSIFICATION

Machine learning has fundamentally transformed the landscape of cyber threat classification by enabling automated systems to learn complex patterns from large datasets and make accurate predictions about the nature and severity of potential threats without relying on manually crafted rules or signatures.

Unlike traditional signature-based detection methods, which can only identify known threats, machine learning models are capable of generalizing from historical data to detect novel and previously unseen attack patterns, making them significantly more effective in addressing the rapidly evolving threat landscape.

Supervised, unsupervised, and semi-supervised learning paradigms have all been extensively explored in the context of cyber security, each offering distinct advantages and trade-offs depending on the availability of labeled data and the complexity of the classification task. Supervised learning approaches represent the most widely adopted category of machine learning techniques for cyber threat classification, leveraging labeled

datasets to train models that can accurately distinguish between benign and malicious content.

Classical supervised classifiers such as Naïve Bayes, Logistic Regression, Support Vector Machines, Decision Trees, and Random Forest have been extensively applied to tasks such as malware detection, intrusion detection, phishing URL classification, and spam filtering.

Deep learning models have further advanced the state of threat classification by enabling the automatic extraction of hierarchical feature representations from raw data, eliminating the need for manual feature engineering.

Recurrent Neural Networks and Long Short-Term Memory networks have proven particularly effective for analyzing sequential and time-series data such as network traffic logs, user behavior patterns, and textual threat communications, where temporal dependencies play a critical role in accurate classification.

CHAPTER 3

METHODOLOGY AND PROPOSED WORK

3.1 THREAT INTELLIGENCE PIPELINE ARCHITECTURE

The Threat Intelligence Pipeline Architecture forms the structural backbone of the proposed Threat Vision system, defining the end-to-end flow of data from initial collection through preprocessing, analysis, risk assessment, and final alert generation.

The architecture is designed as a multi-layered, modular framework that integrates web technologies, backend processing engines, Dark Web access mechanisms, machine learning classification models, and IoT-based physical alert systems into a cohesive and scalable cyber security solution.

Each layer of the pipeline is responsible for a specific set of operations and communicates with adjacent layers through well-defined interfaces, ensuring that data flows seamlessly from one stage to the next while maintaining the integrity, accuracy, and timeliness of the threat intelligence produced at each step. The design of the Threat Intelligence Pipeline Architecture incorporates a comprehensive set of security and privacy considerations that ensure the responsible and legally complaint operation of

the system throughout all stages of the data collection, processing, and analysis workflow. Since the system analyzes sensitive Dark Web content, it operates within a controlled and isolated environment to prevent unauthorized access or exposure of threat data.

3.2 DATA COLLECTION LAYER

The first and foundational layer of the pipeline is the data collection module, which is responsible for continuously and autonomously gathering threat related content from both surface web and Dark Web sources. For surface web URLs, the system employs standard HTTP request mechanisms to fetch webpage content directly, while for Dark Web .onion domains, all traffic is routed through the Tor network using a SOCKS5 proxy configuration running on localhost port 9050, ensuring anonymous and secure access to hidden services. The data collection layer is designed to handle a wide variety of source types including underground forums, darknet marketplaces, paste sites, leak platforms, and encrypted communication channels, providing comprehensive coverage of the threat landscape across both visible and hidden layers of the internet.

3.3 PREPROCESSING AND CLEANING LAYER

Once raw content has been successfully retrieved from the target sources, it is passed to the preprocessing and cleaning layer, which

transforms the raw HTML and textual data into a clean, structured format suitable for downstream analysis. This layer employs HTML parsing libraries such as BeautifulSoup to extract relevant textual content from the surrounding markup, removing scripts, styles, navigation elements, and other non informative components that do not contribute to threat analysis. Regular expression-based cleaning techniques are then applied to remove special characters, excess whitespace, encoded strings, and HTML remnants from the extracted text, producing a normalized and noise-free representation of the original content. Text truncation mechanisms are also implemented at this stage to limit the length of processed documents and optimize system performance, preventing memory overload during the analysis of particularly large or complex web pages.

3.4 FEATURE EXTRACTION AND NLP LAYER

The cleaned and normalized text is subsequently passed to the feature extraction and NLP layer, which applies a range of natural language processing techniques to identify and extract meaningful threat-related information from the preprocessed content. This layer performs tokenization, stop word removal, and normalization to prepare the text for further analysis, followed by the application of keyword extraction techniques such as TF IDF and RAKE to identify the most informative and relevant terms within each document. Named entity recognition is employed to detect specific threat-related entities such as malware names, vulnerability identifiers, attack tools, and targeted organizations, while contextual

semantic analysis techniques are used to identify intent-related phrases and patterns that may indicate coordinated malicious activity. The extracted features are then combined into structured numerical representations that serve as the input to the machine learning classification module, capturing the linguistic, contextual, and structural characteristics of the analyzed content in a format that enables accurate and efficient automated threat classification.

3.5 THREAT COLLECTION AND RISK ASSESSMENT LAYER

The threat classification and risk assessment layer represents the analytical core of the pipeline, applying machine learning models to the extracted feature representations to categorize detected content into relevant threat categories and assign severity scores based on the nature and potential impact of identified threats. In the current implementation of the Threat Vision system, a rule-based classification logic is employed to assign threat levels of Safe, Warning, or Danger based on the presence and frequency of predefined threat-related keywords and patterns within the analyzed content, with a cumulative risk score calculated for each document based on the number and severity of detected indicators. The risk assessment module evaluates multiple factors including threat type, frequency of occurrence, source credibility, and potential damage to targeted systems to produce a comprehensive severity score that enables effective prioritization of identified threats, ensuring that critical and high-impact threats receive

immediate attention while lower-risk activities are monitored for further developments.

3.6 TOR PROXY AND SOCKS5 CONFIGURATION

The Tor (The Onion Router) network forms the critical infrastructure through which the Threat Vision system gains anonymous and secure access to Dark Web sources, enabling the collection of threat-related intelligence from hidden services and .onion domains that are inaccessible through conventional internet connections.

Tor operates by routing all network traffic through a series of volunteer operated relay nodes, each of which decrypts a single layer of encryption to reveal only the address of the next node in the chain, ensuring that no single relay possesses complete knowledge of both the origin and destination of the transmitted data. This layered encryption mechanism provides strong anonymity and privacy protection by making it extremely difficult for external entities to trace user identities, monitor communication paths, or identify the source and destination of internet traffic.

This layered encryption mechanism, which gives the network its name, provides a high degree of anonymity and privacy for both users and services operating within the Tor ecosystem, making it the preferred infrastructure for hosting underground forums, darknet marketplaces, and leak platforms where cybercriminals coordinate and execute their activities

away from the surveillance of law enforcement and conventional cyber security systems.

SOCKS5 is a widely used internet protocol that facilitates the routing of network packets between a client and a server through a designated proxy server, supporting a broad range of traffic types including TCP and UDP connections and providing an additional layer of abstraction between the application and the underlying network infrastructure.

3.7 DATA SCRAPING AND REGULAR EXPRESSION CLEANING

Data scraping and regular expression cleaning constitute two of the most fundamental and technically demanding components of the Threat Vision threat intelligence pipeline, collectively responsible for retrieving raw content from target web sources and transforming it into clean, structured, and analytically meaningful text that can be effectively processed by the downstream natural language processing and machine learning modules.

The quality and reliability of the threat intelligence produced by the system is directly dependent on the effectiveness of these two operations, as poorly scraped or inadequately cleaned data will inevitably introduce noise, inconsistencies, and irrelevant content into the analysis pipeline, reducing classification accuracy and compromising the overall integrity of the threat detection process.

The web scraping module of the Threat Vision system is implemented using Python-based HTTP request libraries that establish connections with target URLs and retrieve the raw HTML content of the corresponding web pages for further processing. For surface web URLs, standard HTTP GET requests are made directly to the target server using the Requests library, which handles connection management, response parsing, and error handling in a straightforward and efficient manner.

3.8 MACHINE LEARNING CLASSIFICATION LOGIC

The classification framework of the Threat Vision system employs a multi-stage evaluation process that systematically assesses the analyzed content against a comprehensive set of threat indicators, progressively building a cumulative risk score that reflects the overall threat level represented by the examined document. The framework is structured around three primary classification categories that provide a clear and intuitive representation of the assessed threat level, enabling security teams to rapidly understand the nature and severity of detected threats and take appropriate mitigation actions without requiring deep technical expertise in cyber security analysis. This classification approach improves the efficiency of threat detection by simplifying complex cyber security analysis into understandable risk levels. It also enhances decision-making capabilities by enabling faster identification, prioritization, and response to potential cyber threats detected from analyzed URLs or documents. Table 3.1 illustrates the Alert Mechanism of proposed system.

Table 3.1 Alert Mechanism

Status	LED	Buzzer	LCD
Safe	Green	OFF	SAFE
Warning	Yellow	OFF	WARNING
Danger	Red	ON	DANGER

Safe — The analyzed content shows minimal or no threat indicators, with a low-risk score indicating that the URL or document is generally safe for access and processing without immediate security concerns.

Warning — The analyzed content contains suspicious patterns or moderate threat-related indicators, suggesting possible malicious activity and requiring further investigation and security attention.

Danger — The analyzed content shows strong evidence of malicious activity, including significant threat indicators and attack-related patterns that represent a high security risk.

CHAPTER 4

REQUIREMENTS SPECIFICATIONS

4.1 HARDWARE SPECIFICATIONS

The hardware components used in this system are designed to provide real time alert mechanisms by integrating IoT devices with cyber security detection. The core component is the ESP32, which acts as the main controller of the system. It is responsible for processing data received from the backend server, managing Wi-Fi communication, and controlling all connected hardware components. The ESP32 is preferred due to its built-in Wi-Fi and Bluetooth capabilities, low power consumption, and high processing efficiency.

For user notification, an 16x2 LCD Display is used to display real-time threat status such as “SAFE,” “WARNING,” or “DANGER.” This provides a clear visual output for users. Along with the LCD, LED indicators (green, yellow, red) are used to represent different threat levels, and a buzzer is included to provide an audio alert in high-risk situations.

Additional supporting components include a breadboard or PCB for circuit assembly, jumper wires for connections, and resistors to regulate current flow. Together, these components form a complete hardware setup

that ensures immediate and effective physical alerting in response to detected cyber threats. Table 4.1 illustrates the hardware requirements of the proposed system.

Table 4.1 Hardware Cost Estimation

Component	Quantity	Approximate Cost (INR)
Breadboard	1	80–150
Jumper Wires	1 set	80–150
USB Cable	1	50–150
Project Box / Enclosure	1	150–400
Zero PCB / Perf Board	1	30–80
ESP32 / ESP8266 Board	1	300–600
16x2 LCD Display with I2C	1	150–250
LEDs (Red, Yellow, Green)	3	15–30
Resistors	3–5	10–20
Buzzer	1	30–80

4.2 SOFTWARE SPECIFICATIONS

The software components of the system are responsible for handling data processing, threat analysis, communication, and integration with hardware modules. The primary programming language used is Python, which provides flexibility, simplicity, and a wide range of libraries for web development, data processing, and cyber security applications. Python is used to implement backend logic, perform data scraping, process text, and execute threat detection algorithms.

The backend framework used is FastAPI, a modern and high-performance web framework for building APIs. FastAPI is responsible for handling HTTP requests, processing user inputs, and returning analysis results. It provides endpoints such as URL analysis, report generation, and real-time status updates for IoT devices. Table 4.2 illustrates the Software requirements of the proposed system.

Table 4.2 Software Cost Estimation

Software / Tool	Purpose	Cost
Python	Backend development	Free
FastAPI	API framework	Free
Arduino IDE	IoT programming	Free
Tor Browser / Tor Service	Dark web access	Free
HTML, CSS, JavaScript	Frontend	Free
Chrome Extension Platform	Browser plugin	Free
VS Code	Development IDE	Free
Pandas / BeautifulSoup	Processing libraries	Free

Additional supporting libraries include BeautifulSoup for HTML parsing and data extraction, Regular Expressions (re module) for text cleaning and pattern matching, and Pandas for generating CSV reports. Frontend technologies such as HTML, CSS, and JavaScript are used to create a user-friendly interface. Together, these software components form a

complete system capable of real-time threat detection, dark web analysis, and IoT-based alert integration.

4.3 DATABASE SCHEMA AND API ENDPOINTS

The system uses a simple and flexible database structure to store analysis results and support future scalability. The database schema is designed to capture essential information related to each scanned URL, enabling efficient tracking, reporting, and retrieval of threat data.

The key fields in the schema include URL, which stores the input link provided by the user; Status, which represents the threat level (Safe, Warning, or Danger); Score, which indicates the calculated risk value based on analysis; and Timestamp, which records the date and time of the scan. This structured format allows easy storage and retrieval of historical data for monitoring trends and generating reports.

The schema can be implemented using lightweight databases such as SQLite for small-scale systems or extended to more robust solutions like MySQL or MongoDB for large-scale deployments. The system also provides several API endpoints developed using FastAPI to handle communication between the frontend, backend, and IoT devices. Table 4.3 illustrates the API Endpoints of the proposed system.

Table 4.3 API Endpoints

Endpoint	Method	Description
/analyze	POST	Analyze URL
/analyze_csv	POST	Download report
/get	GET	IoT status
/health	GET	Server check
/dashboard	GET	View results

4.4 SYSTEM FLOWCHART AND CIRCUIT DIAGRAM

The system flowchart represents the step-by-step working process of the proposed cyber security and IoT-based alert system. The process begins when the user enters a URL through the web interface or browser extension.

This input is sent to the backend server developed using FastAPI, where the request is processed. The system then checks whether the URL belongs to the surface web or a dark web (.onion) domain. If it is a dark web link, the request is routed through the Tor proxy for secure access. After fetching the webpage content, the data is cleaned using parsing and preprocessing techniques. The cleaned data is then passed to the threat analysis engine, where keywords and patterns are analyzed to calculate a risk score. Based on this score, the system classifies the threat level as Safe, Warning, or Danger.

The result is displayed on the dashboard, stored in the database, and simultaneously sent to the IoT module. Finally, the ESP32 device receives the status and triggers the appropriate hardware response such as LED indication, buzzer alert, and LCD display output. Figure 4.1 illustrates the Hardware Circuit for Smart Threat Detection System of proposed system.

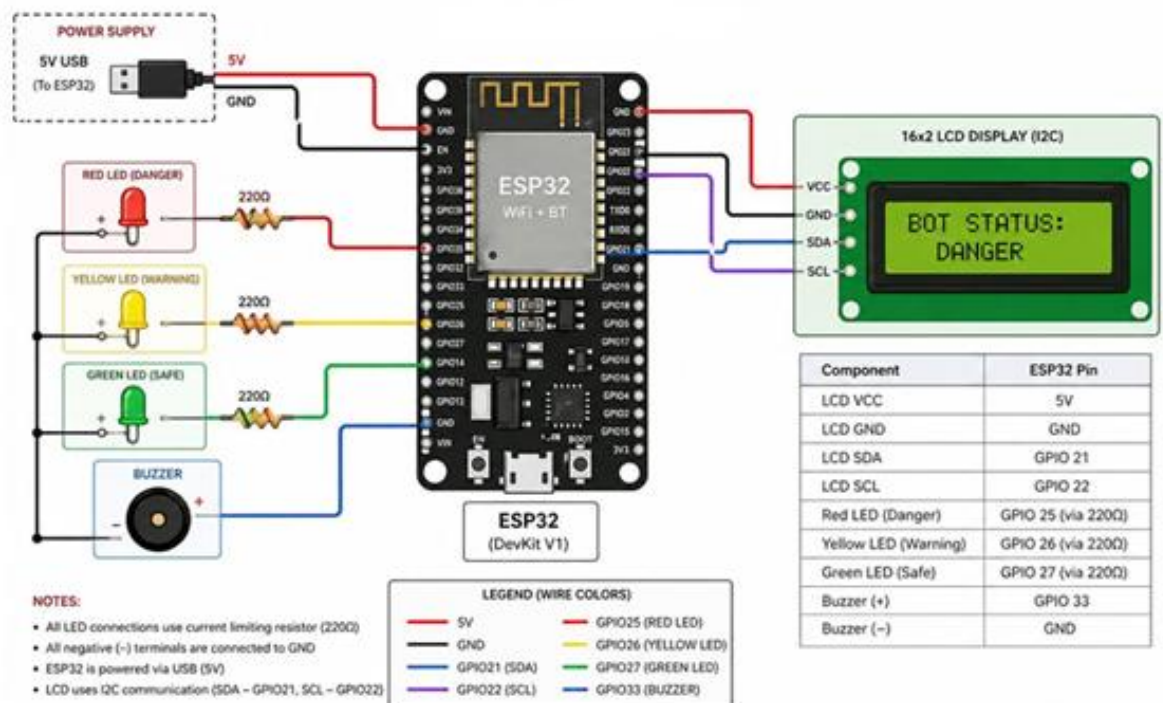


Figure 4.1 Hardware Circuit for Smart Threat Detection System

4.5 NETWORKS AND COMMUNICATION ARCHITECTURE

The network and communication architecture of the proposed system is designed to enable seamless interaction between the user interface, backend server, dark web integration module, and IoT-based hardware components.

The system follows a client-server architecture where the user interacts with a web interface to submit URLs for analysis. These requests are sent to the backend server developed using FastAPI, which processes the input and performs threat detection. Once a URL is received, the backend determines whether it belongs to the surface web or the dark web. For dark web links, communication is routed through the Tor network using SOCKS5 proxy configuration, ensuring secure and anonymous data retrieval.

The communication between the backend and the IoT device is established over a Wi-Fi network. The ESP32 periodically sends HTTP GET requests to the backend API endpoint (such as /get) to retrieve the latest threat status. Based on the received response, the ESP32 activates the corresponding hardware components, including LEDs, buzzer, and LCD display.

4.6 SECURITY AND PRIVACY CONSIDERATIONS

Security and privacy are critical aspects of the proposed system, especially due to its involvement in handling sensitive data and analyzing both surface web and dark web content. To ensure secure communication, the system implements encrypted data transmission using HTTPS, which protects information from interception and unauthorized access during network communication.

The system also focuses on data privacy protection by minimizing the storage of sensitive information. User inputs such as URLs and analysis results are stored only when necessary and can be anonymized to prevent exposure of personal or confidential data. Access control mechanisms can be applied to the database to ensure that only authorized personnel can view or modify stored information.

Another important consideration is secure dark web interaction. Since the system accesses hidden networks through Tor, it is essential to operate in a controlled and isolated environment to avoid exposure to malicious content. Proper filtering and sandboxing techniques can be used to prevent harmful data from affecting the system.

Furthermore, logging and monitoring mechanisms are implemented to track system activities and detect suspicious behavior. This helps in identifying potential security breaches and supports incident response. Regular updates, patch management, and vulnerability assessments are also essential to maintain system integrity.

CHAPTER 5

MODULES AND IMPLEMENTATION

5.1 WEB INTERFACE AND DASHBOARD DEVELOPMENT

The interface and dashboard serve as the primary interaction layer between the user and the system, providing a simple, responsive, and user-friendly environment for cyber security analysis. The interface is developed using standard web technologies such as HTML, CSS, and JavaScript, ensuring compatibility across different devices and browsers. It allows users to input URLs, initiate threat analysis, and view results in real time through an intuitive layout.

The backend communication is handled using FastAPI, which processes user requests and returns results in a structured format such as JSON. The frontend uses asynchronous techniques (such as fetch API or AJAX) to send requests to the backend without reloading the page, ensuring a smooth and dynamic user experience.

The dashboard includes several key components such as a URL input field, an analyze button, a results display section, and an option to download reports in CSV format. The threat status is visually represented using color

indicators— green for safe, yellow for warning, and red for danger—making it easy for users to quickly understand the system output. The dashboard provides a user-friendly interface for efficient real-time cyber threat monitoring and analysis.

The dashboard also supports real-time updates, where the results of threat analysis are displayed instantly without requiring page refresh. This is achieved through asynchronous communication with the backend using FastAPI, ensuring smooth and fast interaction. Figure 5.1 illustrates the Threat Vision System Dashboard Interface of proposed system.

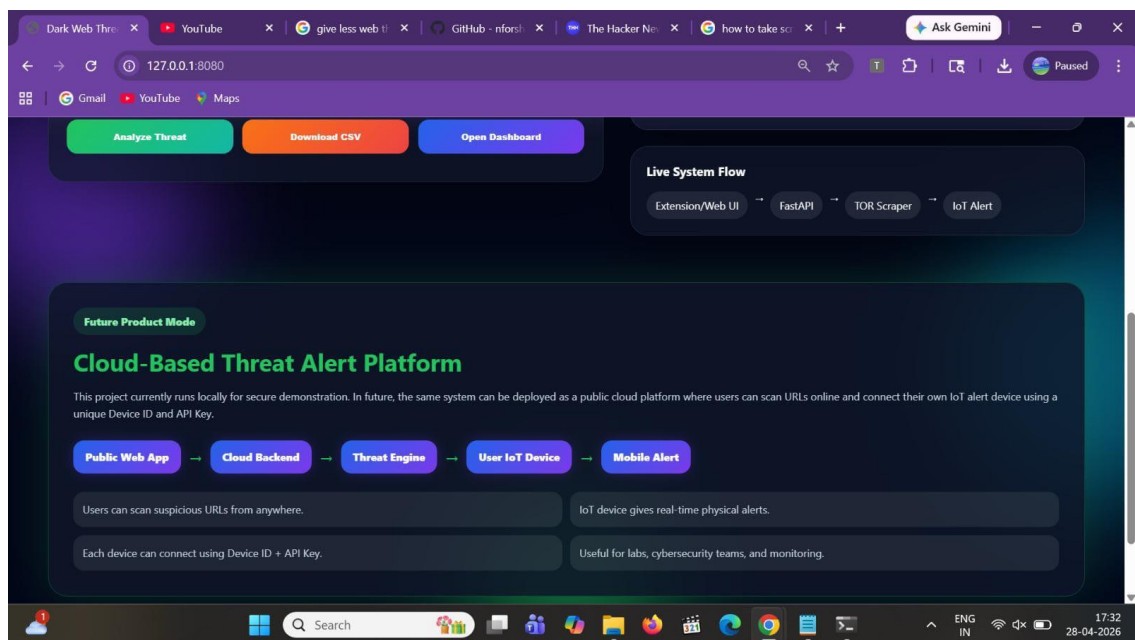


Figure 5.1 Threat Vision System Dashboard Interface

5.2 DARK WEB SCRAPING MODULE

The dark web scraping module is a crucial component of the system, responsible for extracting relevant data from hidden web sources that are not accessible through standard browsers. These sources typically operate on anonymized networks such as Tor, which enables secure and anonymous communication. To access such content, the system is configured to route requests through a SOCKS5 proxy (e.g., socks5h://127.0.0.1:9050), allowing it to connect to .onion domains.

Once a connection is established, the system sends HTTP requests to retrieve the HTML content of the target webpage. The fetched content often includes unnecessary elements such as scripts, advertisements, and navigation components. To extract meaningful information, the system uses parsing tools like BeautifulSoup to process the HTML structure and isolate the relevant textual data.

After extraction, the data undergoes cleaning and preprocessing using techniques such as regular expressions to remove unwanted characters, symbols, and formatting inconsistencies. This step ensures that the data is structured and suitable for further analysis.

The cleaned text is then passed to the threat analysis engine, where it is evaluated for suspicious keywords, patterns, and indicators of cyber threats. The module also includes mechanisms for handling challenges

associated with dark web scraping, such as slow response times, unstable connections, and dynamic content. Error handling and timeout management are implemented to ensure system stability during data retrieval.

5.3 THREAT ANALYSIS ENGINE

The Threat Analysis Engine is the core component of the system responsible for evaluating extracted data and determining the level of cyber security risk. It processes the cleaned text obtained from both surface web and dark web sources and applies analytical techniques to identify potential threats. The engine is designed using a combination of rule-based logic and provisions for integrating artificial intelligence models to enhance detection accuracy.

The analysis begins with text processing, where the input data is structured and prepared for evaluation. The system then performs keyword detection, identifying terms commonly associated with cyber threats such as malware, phishing, ransomware, exploit, and hacking. In addition to keywords, the engine uses pattern recognition techniques to detect suspicious elements such as unusual URLs, encoded strings, or abnormal content structures. Figure 5.2 illustrates the Threat Analysis Flowchart of proposed system.

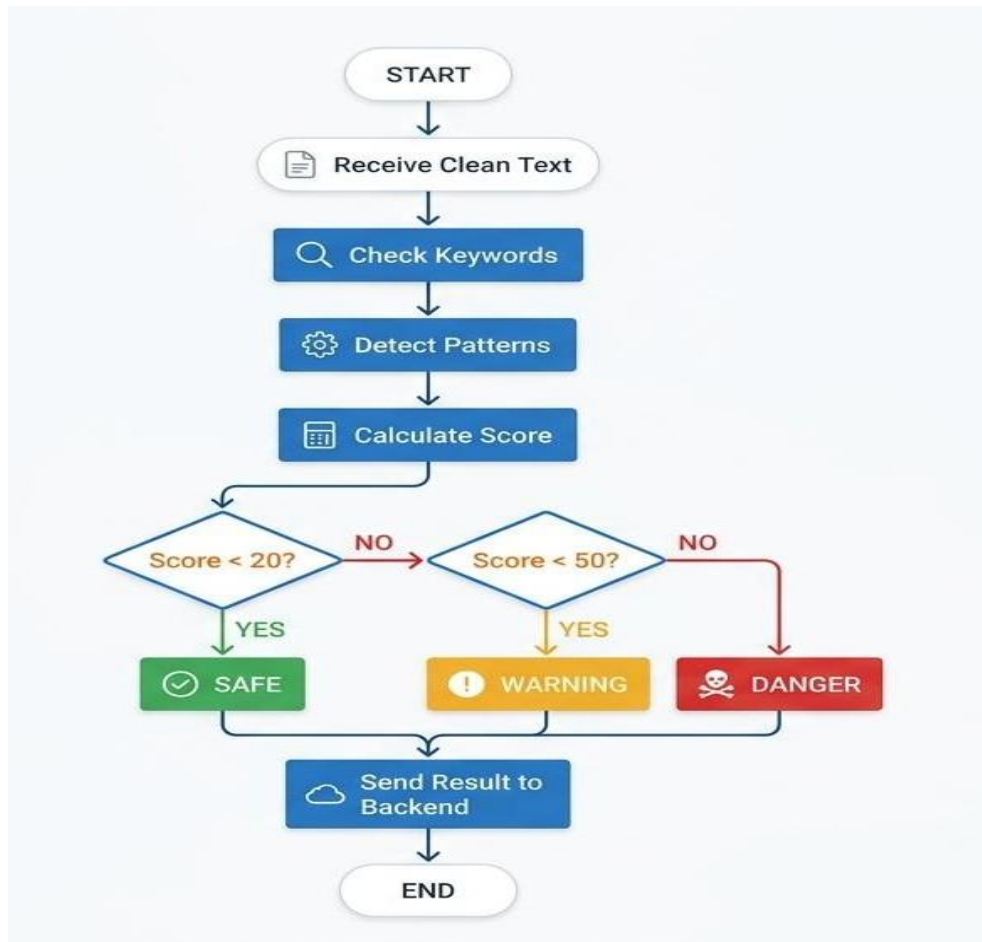


Figure 5.2 Threat Analysis Flowchart

The analysis begins with text processing, where the input data is structured and prepared for evaluation. The system then performs keyword detection, identifying terms commonly associated with cyber threats such as malware, phishing, ransomware, exploit, and hacking. In addition to keywords, the engine uses pattern recognition techniques to detect suspicious elements such as unusual URLs, encoded strings, or abnormal content structures.

5.4 IOT HARDWARE ALERT AND NOTIFICATION MODULE

The IoT Hardware Alert and Notification Module is responsible for converting digital threat detection results into real-time physical alerts, ensuring immediate user awareness even when the system interface is not actively monitored. This module is built around the ESP32, which acts as the central controller for communication and hardware interaction.

The ESP32 connects to the backend server over Wi-Fi and continuously retrieves the latest threat status using API requests. Based on the received response (Safe, Warning, or Danger), it activates the corresponding output devices. Visual alerts are provided using LED indicators—green for safe, yellow for warning, and red for danger—while an audio alert is generated using a buzzer in high-risk situations. Additionally, a 16x2 LCD Display is used to display the current threat status, providing clear and immediate feedback to the user.

The working process of the module involves connecting the ESP32 to a Wi-Fi network, sending periodic requests to the backend API (such as /get), receiving the analyzed threat status, and triggering the appropriate hardware response. This continuous communication ensures real-time updates and rapid response to potential threats.

5.5 BROWSER EXTENSION DEVELOPMENT

The browser extension is an important component of the system that provides users with a convenient and real-time method to analyze URLs directly from their web browser. It eliminates the need to manually visit the web application, allowing users to quickly scan websites for potential threats while browsing. The extension is typically developed using web technologies such as HTML, CSS, and JavaScript, ensuring compatibility with popular browsers like Chrome and Firefox. Figure 5.3 illustrates the Threat Analyzer Browser Extension Interface of proposed system.

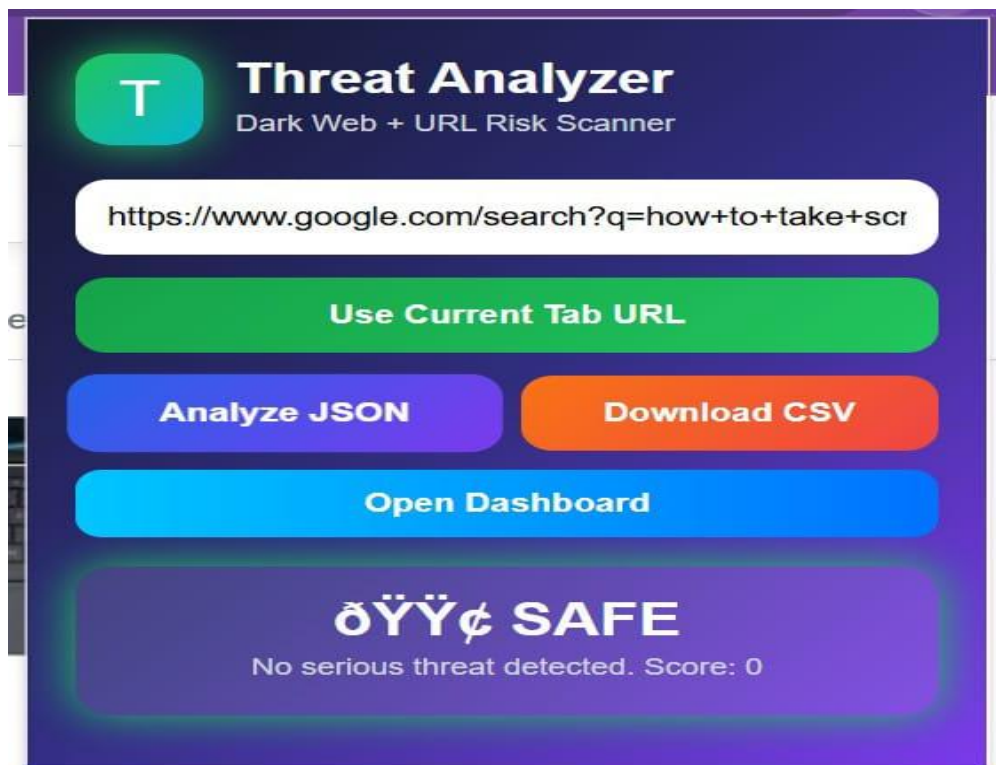


Figure 5.3 Threat Analyzer Browser Extension Interface

The user interface of the extension is designed to be simple and intuitive, typically including features such as a scan button, result display, and visual indicators using color codes (green, yellow, red). This ensures that users can quickly understand the security status of a website without technical complexity.

5.6 CSV REPORT GENERATION MODULE

The CSV Report Generation Module is responsible for creating structured and downloadable reports of the analyzed data, allowing users to store, review, and share threat analysis results efficiently. This module converts the processed data into a tabular format and exports it as a CSV (Comma-Separated Values) file, which can be easily opened using spreadsheet applications such as Excel. Figure 5.4 illustrates the Input Interface for Standard and onion URL Analysis of proposed system.

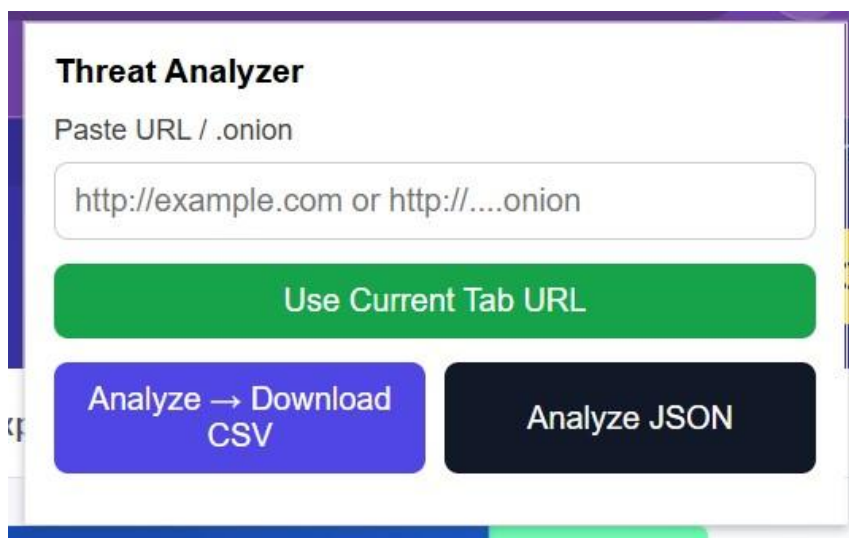
The image shows a web-based interface titled "Threat Analyzer". Below the title is a text input field with the placeholder text "Paste URL / .onion". Inside the input field, the text "http://example.com or http://...onion" is visible. Below the input field is a green button labeled "Use Current Tab URL". At the bottom of the interface, there are two buttons: a blue button labeled "Analyze → Download CSV" and a dark blue button labeled "Analyze JSON".

Figure 5.4 Input Interface for Standard and onion URL Analysis

The module collects essential information from the analysis process, including the input URL, detected threat status (Safe, Warning, or Danger), calculated risk score, identified keywords, and timestamp of the analysis. This data is organized into rows and columns, making it easy to read and interpret. The report provides a clear overview of all analyzed entries, which is useful for documentation, auditing, and further investigation.

5.7 TELEGRAM NOTIFICATION INTEGRATION

The Telegram Notification Integration module enables the system to send real-time alerts directly to users through the messaging platform Telegram. This feature ensures that users receive instant notifications about detected threats, even when they are not actively monitoring the dashboard or IoT hardware alerts.

The integration is implemented using the Telegram Bot API, where a bot is created and configured to communicate with the backend system. The bot is assigned a unique API token, which is used by the backend (developed in FastAPI) to send messages programmatically. When a threat is detected, the system triggers a request to the Telegram API, delivering a message that includes details such as the analyzed URL, threat level (Safe, Warning, or Danger), and risk score.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 PERFORMANCE AND EVALUATION OF AI MODELS

The performance evaluation of AI models is a critical step in assessing the effectiveness and reliability of the threat detection system. It ensures that the implemented models can accurately identify cyber threats while minimizing errors. The evaluation process involves testing the model on both training and unseen datasets to measure its real-world performance.

One of the key metrics used is accuracy, which indicates the overall correctness of the model in classifying data as Safe, Warning, or Danger. However, accuracy alone is not sufficient, especially in cyber security applications. Therefore, additional metrics such as precision, recall, and F1-score are used. Precision measures how many detected threats are actually correct, while recall evaluates how many actual threats are successfully identified by the model. The F1-score provides a balance between precision and recall, giving a more comprehensive performance measure.

Another important aspect is the confusion matrix, which provides a detailed breakdown of correct and incorrect predictions. It helps in identifying false positives (normal data classified as threats) and false negatives (threats not detected), both of which are critical in cyber security

systems. The system also evaluates model efficiency and response time, ensuring that threat detection is performed in real time without significant delay. This is particularly important for applications involving live data processing and IoT based alerts.

6.2 REAL-TIME RESPONSE LATENCY ANALYSIS

Real-time response latency analysis refers to the measurement and evaluation of the time delay between the detection of a cyber threat and the execution of an automated or system-generated response. In modern cyber security systems, especially in domains like dark web threat intelligence, minimizing latency is critical to prevent data breaches, cyberattacks, and unauthorized access.

This analysis focuses on tracking the complete lifecycle of threat handling, starting from data collection (such as monitoring dark web forums, leaked databases, and suspicious network activity) to processing, detection using AI/ML models, alert generation, and final response execution (such as blocking an IP, notifying administrators, or isolating a system).

The overall performance of a cyber security system depends heavily on reducing each of these latency stages. High latency can lead to delayed responses, increasing the risk of data leakage or system compromise. On the other hand, optimized low-latency systems ensure real-time protection and immediate threat mitigation.

It focuses on measuring the time taken for the system to detect a threat, process the information, and execute an appropriate response. The main goal is to ensure that threats are handled in the shortest possible time to minimize damage and prevent security breaches.

6.3 COST ESTIMATION AND PROJECT PLANNING(COCOMO)

COCOMO stands for Constructive Cost Model. It is a software cost estimation model used to estimate the effort, development time, and manpower required for a software project. It was introduced by Barry Boehm and is widely used in software engineering project planning.

Cost estimation is an important part of software engineering because it helps in understanding the resources, time, effort, and budget required to complete a project successfully. In this project, Threat Vision – Dark Web and IoT Threat Intelligence System, cost estimation is required because the project includes multiple components such as software development, backend API implementation, web interface design, browser extension development, IoT hardware integration, testing, and report preparation.

The general formula for effort estimation is:

$$\text{Effort} = a \times (\text{KLOC})^b$$

The development time is calculated using: $\text{Development Time} = c \times (\text{Effort})^d$ Where Effort = Person-months

KLOC = Thousand lines of code

a, b, c, d = Constants based on project type COCOMO classifies software projects into three main categories:

Table 6.1 illustrates the COCOMO Project Categories and Descriptions of proposed system

Table 6.1 COCOMO Project Categories and Descriptions

Project Type	Description
Organic	Small projects developed by a small team with familiar technology
Semi-detached	Medium complexity projects with mixed experience
Embedded	Complex projects with strong hardware/software constraints

The Threat Vision project involves both software and hardware components, but the hardware part is not deeply complex. The project includes web development, backend APIs, IoT communication, and dark web analysis. Therefore, it can be classified as a semi-detached project.

6.4 THREAT CLASSIFICATION ACCURACY AND CONFUSION MATRIX

The evaluation of threat classification accuracy and the analysis of confusion matrices represent essential components of the performance

assessment framework for the Threat Vision system, providing a comprehensive and quantitative understanding of the system's ability to correctly identify and categorize cyber threats across the three defined risk levels of Safe, Warning, and Danger.

A rigorous and systematic evaluation of classification performance is critical for establishing the reliability and practical utility of the threat detection pipeline in real-world cyber security operations, as inaccurate classifications can have serious consequences including missed detections of genuine threats that allow malicious activity to proceed unimpeded and false positive alerts that overwhelm security teams with spurious notifications and erode confidence in the system's outputs.

Accuracy measures the overall proportion of correctly classified instances out of the total number of analyzed URLs, providing a high-level indicator of the system's general classification reliability across all three threat categories combined.

Precision evaluates the reliability of positive threat detections by measuring the proportion of instances classified as belonging to a particular threat category that are genuinely members of that category, providing a quantitative measure of the false positive rate for each classification level. It is widely used to assess model trustworthiness, especially in cyber security systems where incorrect alerts can lead to unnecessary investigations and resource consumption.

Recall measures the system's capability to correctly identify all genuine instances of each threat category within the analyzed dataset, providing a quantitative measure of the false negative rate for each classification level.

6.5 IOT ALERT RESPONSE VALIDATION

The IoT Alert Response Validation constitutes a critical component of the Threat Vision system evaluation framework, providing a systematic and comprehensive assessment of the physical alert delivery mechanism's reliability, responsiveness, and accuracy in correctly translating software-generated threat classifications into immediate and appropriate hardware notification responses.

The IoT alert response validation methodology employed in the Threat Vision evaluation framework is structured around a systematic series of test scenarios that collectively assess all critical aspects of the physical alert delivery mechanism, from the initial WiFi connection establishment and backend polling initiation through the final hardware output activation and LCD display update.

The threat level transition validation tests evaluate the correctness and reliability of the IoT alert mechanism's response to changes in the current threat status, ensuring that the hardware outputs correctly reflect each new threat classification as the backend processes different URLs and updates the current threat status accordingly.

6.6 COMPARISION WITH EXISTING SYSTEMS

The comparative analysis of the Threat Vision system against existing cyber security solutions provides a comprehensive and objective assessment of the distinctive capabilities and competitive advantages that the proposed framework offers relative to the current state of the art in cyber threat detection and alerting technology. Table 6.2 illustrates the Threat Vision Feature Set vs. Conventional Tools of proposed system.

Table 6.2 Threat Vision Feature Set vs Conventional Tools

Feature	Traditional IDS/Firewall	VirusTotal	Dark Web Monitor	Conventional IoTAlert	Proposed ThreatVision
Dark Web URL Analysis	No	No	Yes	No	Yes
IoT Physical Alert Mechanism	No	No	No	Yes	Yes
AI-Driven Threat Classification	Partial	Partial	Partial	No	Yes
Multi-Channel Alert Delivery	No	Partial	Partial	No	Yes
Proactive Threat Intelligence	No	No	Partial	No	Yes

The comparative analysis presented across all four tables collectively demonstrates that the Threat Vision system occupies a unique and distinctive position within the cyber security solution landscape, offering a combination

of Dark Web intelligence, AI-driven threat classification, multi-channel alert delivery, and IoT-based physical notification capabilities that is not currently available in any single existing commercial or open-source cyber security product.

The results of this comparative analysis clearly demonstrate the unique value proposition of the Threat Vision system and identify the specific feature combinations that distinguish it from all existing solutions currently available in the cyber security market.

While enterprise solutions such as Recorded Future and commercial Dark Web monitoring platforms offer more advanced machine learning capabilities and broader data source coverage, they lack the physical IoT alert mechanism, low-cost deployment model, and integrated hardware-software architecture that constitute the most innovative and practically valuable contributions of the Threat Vision framework.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 KEY CONTRIBUTIONS AND OUTCOMES

The Threat Vision project has delivered a range of significant technical contributions and measurable outcomes that collectively advance the current state of the art in proactive cyber security threat detection and physical alert delivery, demonstrating the practical feasibility and operational value of integrating Dark Web intelligence, AI-driven threat analysis, and IoT-based hardware notification capabilities within a single unified and accessible cyber security framework.

The key contributions and outcomes of the project are presented and discussed in detail below, organized according to their primary domain of contribution spanning system architecture, technical innovation, academic knowledge advancement, and practical deployment utility.

7.2 FUTURE ENHANCEMENTS

Although the current system is functional, several improvements can be made to enhance its accuracy, scalability, and usability.

Cloud Deployment - The current system runs locally. In the future, the backend can be deployed on cloud platforms such as AWS, Google Cloud, Azure, Render, or Railway. This will allow users to access the system from anywhere.

Machine Learning-Based Threat Detection - The current version uses rule-based detection. In the future, machine learning models such as LSTM, Random Forest, Support Vector Machine, or transformer based models can be used to improve classification accuracy. ML models can learn from large datasets and identify threats more accurately than manually written rules.

Trusted Domain Whitelist - To reduce false positives, trusted domains such as government, educational, and official websites can be whitelisted. For example: .gov , .edu , .ac.in , .gov.in , .nic.in . This will reduce the chance of safe websites being classified as dangerous.

7.3 SCOPE FOR CLOUD DEPLOYMENT AND COMMERCIALISATION

The Threat Vision system in its current form operates as a locally deployed prototype that has successfully demonstrated the technical feasibility and operational value of integrating Dark Web intelligence, AI-driven threat classification, and IoT-based physical alert delivery within a single unified cyber security framework.

However, the full commercial and operational potential of the proposed system can only be realized through migration to a cloud-based deployment architecture that enables global accessibility, elastic scalability, enterprise-grade reliability, and seamless integration with the broader ecosystem of commercial cyber security platforms and services.

The cloud deployment architecture for the Threat Vision system is envisioned as a containerized microservices framework deployed on a major cloud platform such as Amazon Web Services, Google Cloud Platform, or Microsoft Azure, leveraging the elastic scaling, global distribution, and managed service capabilities of these platforms to deliver the performance, reliability, and accessibility characteristics required for enterprise-grade cyber security operations.

The long-term commercial vision for the Threat Vision system envisions its evolution into a comprehensive and globally recognized proactive cyber security intelligence platform that combines the most advanced capabilities of Dark Web monitoring, artificial intelligence-driven threat analysis, and physical IoT-based alert delivery into a distinctive and highly differentiated product offering that addresses the growing organizational demand for proactive rather than reactive cyber security defense capabilities.

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LIST OF PUBLICATIONS

International Journal

GitHub Link

<https://github.com/PrathikaPR/ml-based-dark-web-threat-classification>

YouTube Link

<https://youtu.be/FQWb1AwqQoQ>